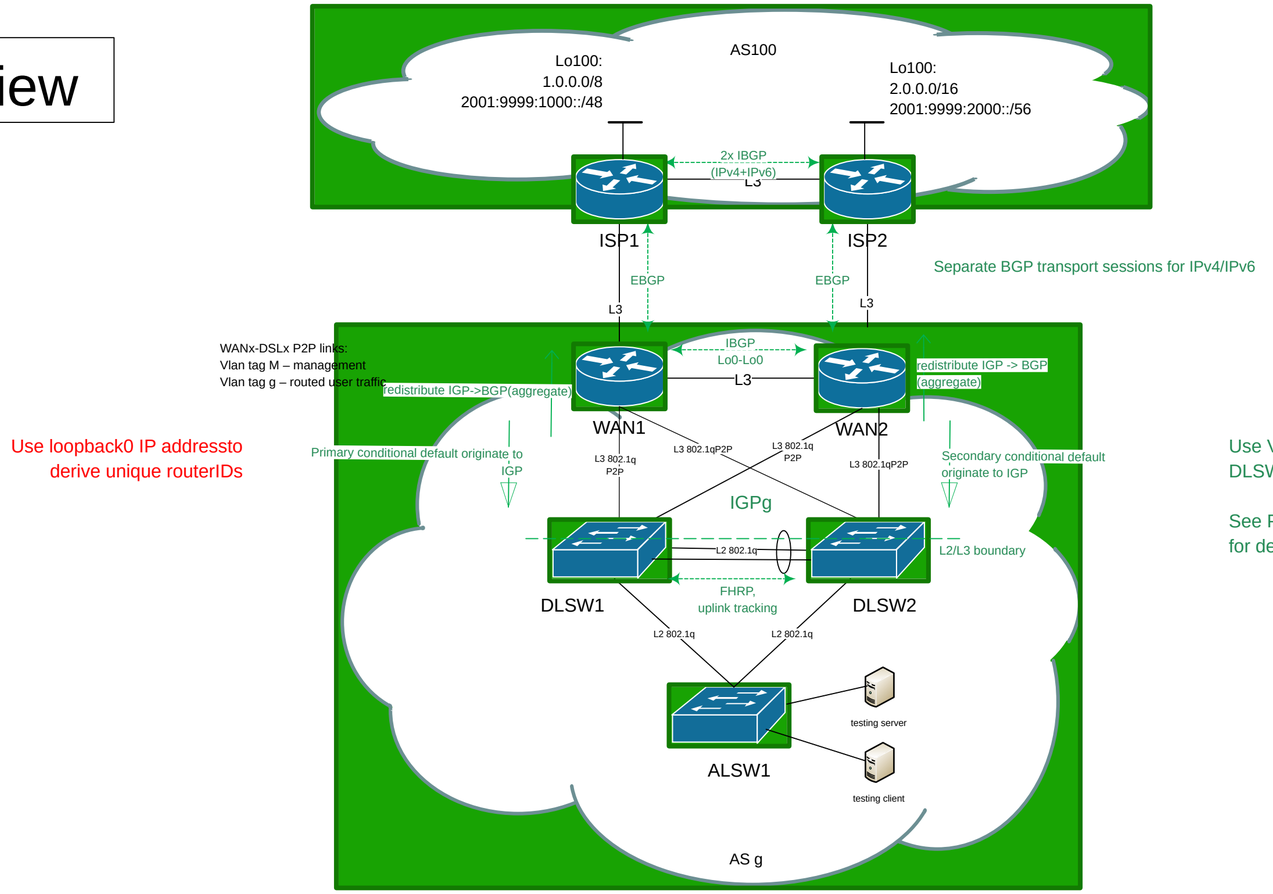
Routed and Switched Networks

Petr Grygárek

# Network Configuration Case Study

In following diagrams, g denotes your group number

# Overview

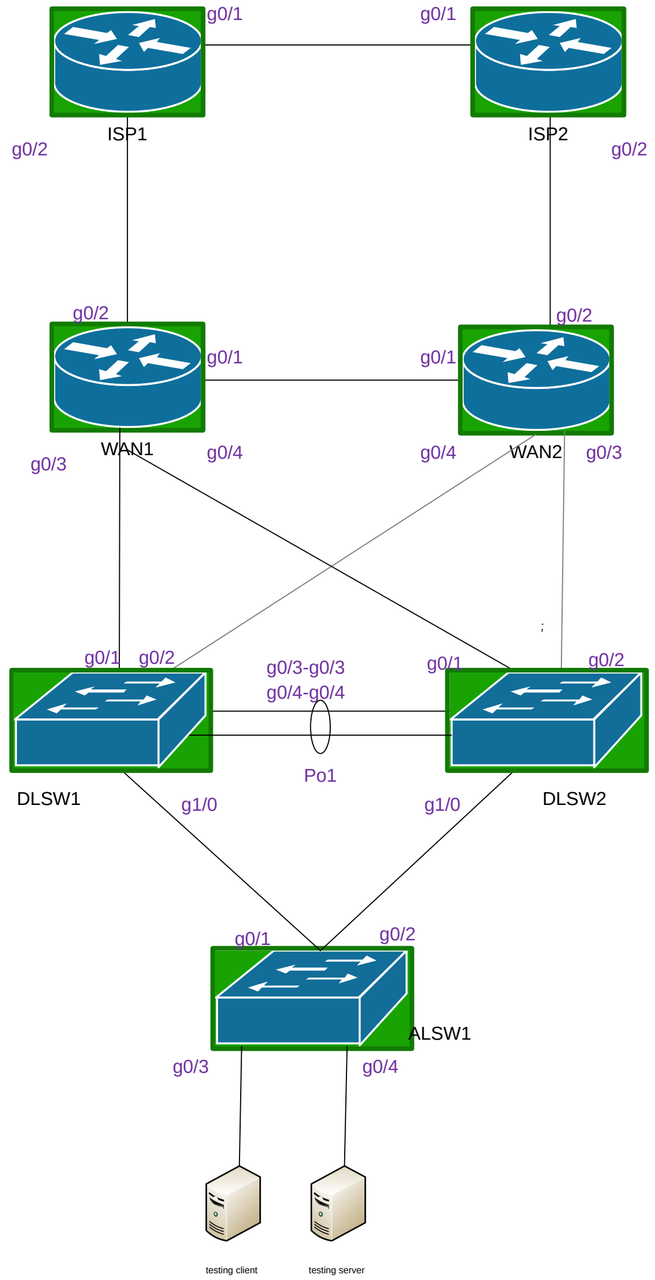



Use loopback0 IP addressto
derive unique routerIDs

Use VLAN tag g0 on IC links between WANx and DLSWx

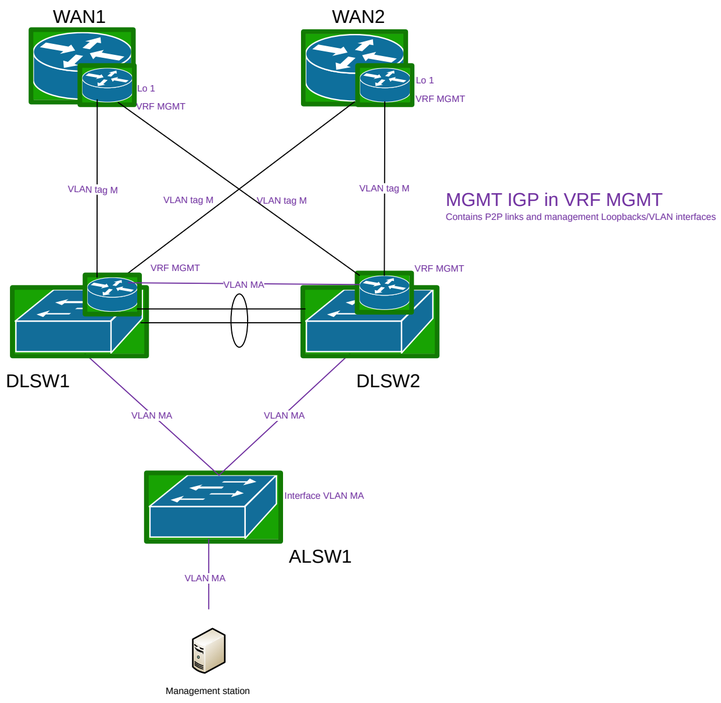
See Parameters sheet of XLS document
for detailed requirements of IGP configuration

Virl

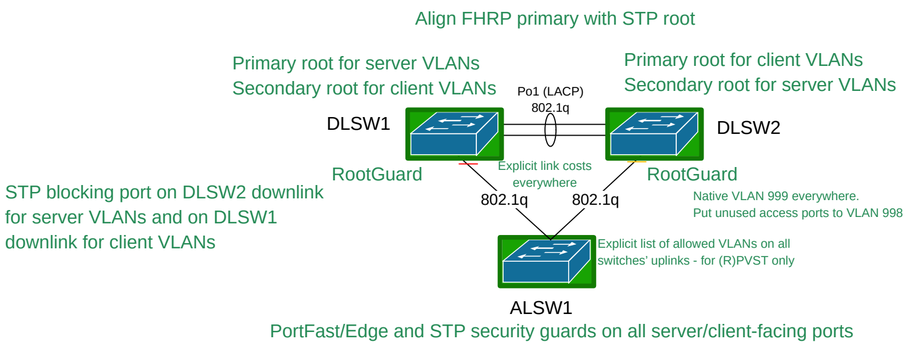

g0/1
g0/1
ISP1
ISP2
g0/2
g0/2
g0/2
g0/2
g0/1
g0/1
WAN1
WAN2
g0/3
g0/4
g0/4
g0/3
;
g0/1
g0/2
g0/3-g0/3
g0/4-g0/4
g0/1
g0/2
Po1
DLSW1
DLSW2
g1/0
g1/0
g0/1
g0/2
ALSW1
g0/3
g0/4
testing client
testing server

# Inband Management

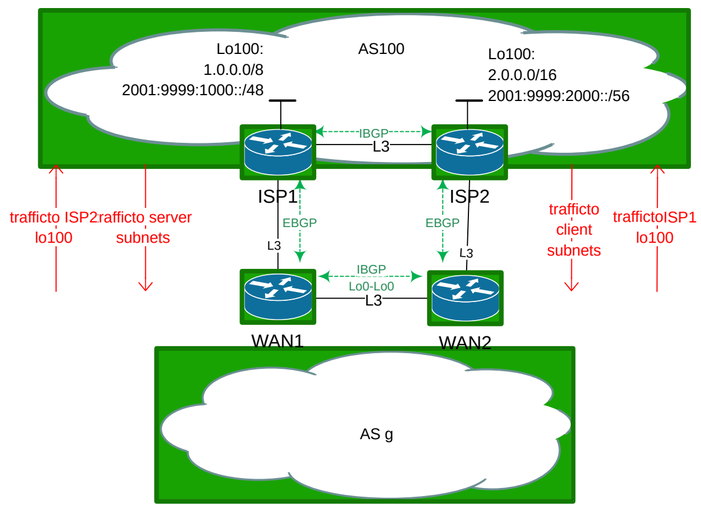
(Management of ISP routers is out of scope of this case study)



# L2 Details


Align FHRP primary with STP root
Primary root for server VLANs
Secondary root for client VLANs
Primary root for client VLANs
Secondary root for server VLANs
Po1 (LACP)
802.1q
DLSW1
DLSW2
RootGuard
RootGuard
Explicit link costs everywhere
802.1q
802.1q
STP blocking port on DLSW2 downlink for server VLANs and on DLSW1 downlink for client VLANs
Native VLAN 999 everywhere.
Put unused access ports to VLAN 998
Explicit list of allowed VLANs on all switches' uplinks - for (R)PVST only
ALSW1
PortFast/Edge and STP security guards on all server/client-facing ports

# BGP


Lo100:
1.0.0.0/8
2001:9999:1000::/48
AS100
Lo100:
2.0.0.0/16
2001:9999:2000::/56
IBGP
L3
ISP1
ISP2
trafficto ISP2
lo100
rafficto server
subnets
EBGP
EBGP
L3
L3
IBGP
Lo0-Lo0
L3
trafficto
client
subnets
trafficto
lo100
ISP1
WAN1
WAN2
AS g